

Ecological Moral Realism: An Alternative Theoretical Framework for Studying Moral Psychology

Eranda Jayawickreme
University of Pennsylvania

Anthony Chemero
Franklin and Marshall College

The authors propose that the empirical study of human moral behavior as undertaken by positive psychologists, psychologists interested in the study of values, and experimental ethicists can be normative at its core, yet also remain truly scientific. To do this, those interested in the empirical study of human moral behavior should consider the concept of affordances, which is central to the Gibsonian ecological psychology tradition, as an approach to studying virtues and strengths. From this perspective, virtues can be better defined as part of moral analogues of affordances (MAAs). Just as the affordance “being-climbable” is a relation between climbing ability and height, so one can similarly define a MAA as an opportunity for moral behavior. Virtues, on this account, would be defined as abilities to behave appropriately in morally relevant situations. If one studies virtues as components of MAAs, virtues are only comprehensible in terms of morally relevant situations. Similarly, morally relevant situations are comprehensible only in terms of abilities to behave appropriately in them, that is, in terms of virtues. We believe that such an approach holds many advantages.

Keywords: affordances, values, virtue, ecological psychology, positive psychology, fact/value distinction

In this paper, we propose that those interested in the empirical study of human moral behavior—including proponents of positive psychology and philosophers interested in experimental ethics—can, as it were, have their cake and eat it too. That is, we propose to explain how the scientific study of moral behavior can be normative at its core, but also truly scientific—achieving results that are replicable, cumulative, and objective. To do so, we will recommend that researchers interested in an empirically informed account of moral behavior adopt some of the methods of Gibsonian ecological psychology, a contemporary psychology that is in many ways a continuation of the pragmatist psychology of a

century ago (Gibson, 1979; James, 1904). There are several advantages to adopting a broadly Gibsonian perspective to the study of moral behavior. First, it provides a novel theory of the nature of moral virtues. Second, it makes both moral virtues and moral behavior empirically tractable. This latter point should make the perspective especially appealing to empirical ethicists, who have thus far limited their research to conceptual issues.

Psychology, Positivism, and “Pure Facts”

Modern psychology has been suspicious of any attempt to incorporate the study of values into psychology, arguing that such study lies outside the purview of science. Considering values when doing science, on this view, would lead to a skewed outlook that would preclude any successful attempt to get at the truth of a particular situation—the traditional claim of science. Tied to this view, however, is a deeper implicit belief that the fundamental goal of the natural sciences—psychology included¹—is to gain insight into what is undeniably true in the

Eranda Jayawickreme, Positive Psychology Center, University of Pennsylvania; Anthony Chemero, Scientific and Philosophical Studies of Mind Program, Franklin and Marshall College.

We thank John Hunter, James Pawelski, Michael Penn, Stephen Schueller, and Barry Schwartz for helpful comments on drafts.

Correspondence concerning this article should be addressed to Eranda Jayawickreme, Positive Psychology Center, University of Pennsylvania, 3701 Market Street, Suite 200, Philadelphia, PA 19104. E-mail: jayawick@sas.upenn.edu

¹ Although one may argue that psychology has yet to figure out what exactly it is studying (see Reed, 1996).

world. This would explain many psychologists' preference for a hard dichotomy between fact and value, which is in many respects a knee-jerk defense of psychology as an "objective" science: that is, as a value-free examination of features in the natural world.

Indeed, this has been the perspective of leading positive psychologists (e.g., Csikszentmihalyi, 2003; Seligman, 2002; Seligman & Csikszentmihalyi, 2001), despite the fact that they claim to be empirically studying moral virtues. Some background on the positive psychology movement may be useful here. While the study of what can be termed "human strengths" has a long, if very checkered, history in psychology, the advent of positive psychology, championed by researchers including Martin Seligman, Mihaly Csikszentmihalyi, and Ed Diener, has meant that this work now stands at the center of contemporary psychological research. This new field has as its goal the creation of "a psychology of positive human functioning . . . that achieves a scientific understanding and effective interventions to build thriving individuals, families and communities" (Seligman, 2002, pp. 7). As Seligman, the field's most prominent researcher, has written, "we will learn how to build the qualities that help individuals and communities not just endure and survive but also flourish" through the evaluation of positive human traits (Seligman, 2002).

Given "positive" psychology's commitment to study strengths and positive traits—which Seligman refers to as "virtues"—it may seem at first sight that the field attempts to challenge that traditional conception of science as engaged in only the objective study of facts in the world. The study of what contributes to a "flourishing" existence would seem obviously value-laden, and one would expect scholars in the new field to address this concern directly. That the positive psychology movement has so far refused to address this concern points, among other things, to the dominance of the current empirical methodological model in psychology. It also points to positive psychology's fear of becoming marginalized within the field in the same manner that the humanistic psychology movement, which both concerned itself with similar questions of optimal human functioning and eschewed traditional scientific practices. In their flagship article, Seligman and Csikszentmihalyi (2000) make much of the fact

that positive psychology is clearly distinguishable from humanistic psychology through positive psychology's commitment to "science": in the face of criticism from humanistic psychologists, Seligman and Csikszentmihalyi reiterated their view that the humanistic tradition was somehow scientifically "lacking" as a tradition in a "Reply to Comments": "We are, unblushingly, scientists first. The work we seek to support and encourage must be nothing less than replicable, cumulative, and objective" (Seligman & Csikszentmihalyi, 2001, p. 89–90). Csikszentmihalyi reiterates this point in another article, claiming that positive psychology is committed to the current scientific paradigm "for better or for worse" (Csikszentmihalyi, 2003, p. 115).

Part of this commitment to the current scientific paradigm "for better or for worse" is the desire that positive psychology remain value-free. Seligman articulates this most clearly in an extended footnote in his book *Authentic Happiness*. He writes that positive psychology is "not a morality or a worldview; it is a description." He "strongly believes" that science is morally neutral (but ethically relevant). Positive Psychology "describes what the pleasant life, the good life, and the meaningful life are. It describes how to get these lives and what the consequences of living them are. It does not prescribe this life for you, nor does it, as a theory, value any one of these lives above the others" (Seligman, 2002, p. 303). At a purely scientific level, Seligman—like most other psychologists—wants to resist all talk of the possible complication of the fact/value distinction, and has repeated his desire to maintain a strong distinction between the two.

However, this desire to refrain from committing the naturalistic fallacy (drawing ethical conclusions from natural facts) has been disputed in many quarters, including the positive psychology fold. Barry Schwartz writes that a richly developed positive psychology cannot avoid offering prescriptions on what a good life entails:

Shoulds imply claims that are prescriptive rather than descriptive, and psychology, as a positive rather than a normative social science, has tried to steer away from *shoulds*. I believe that if psychologists are serious about turning psychology's power to developing a theory of optimal functioning, they can no longer avoid *shoulds*. I think that a richly developed psychology

must do more than teach people how to *do* things—it must do more than teach people effective techniques for what they want out of life. It must also tell them something about *what* they should be trying to get (Schwartz, 2000, p.87).

Outside psychology, it has long been widely accepted among philosophers of science that science cannot be a value-free endeavor in totality and that the concept of science as a completely objective practice is a relic of logical positivism. While positivism has now been largely discredited, it has remained the implicit conceptual foundation for many researchers in psychology (Kendler, 2001).

Moreover, a deeper problem exists concerning the fact/value dichotomy. This distinction, originally nothing more than a practical convenience, was hardened into a dichotomy by David Hume, and subsequently propagated by proponents of logical positivism. For the logical positivists, there were just three kinds of statement: analytic statements, synthetic statements, and nonsense. Analytic statements were taken to be true by definition, or by convention. The paradigm cases of analytic statements are statements of mathematics and logic, and sentences by definition such as “All red dogs are red.” Synthetic statements, on the other hand, are statements of fact, are true or false in virtue of everyday experience (“The Eiffel Tower is taller than any cow”) or the more systematic experience made available using the methods of science (“The atomic number of helium is 2”). Everything else, including aesthetic (“The Eiffel Tower is lovely”) and moral statements (“It was wrong to steal the doughnut”), is nonsense. The positivist position was that scientific theories are constructed of synthetic statements of fact, with analytic math and logic as the foundation, with no room for moral nonsense.

However, the distinction between the analytic and the synthetic, the heart of logical positivism, was comprehensively demolished in the 1950s. First, Quine (1953) showed that the statements of scientific theories are not easily sortable into the analytic, the synthetic and everything else. His means of showing this was to show that no definition of “analytic” was both adequate to account for all the analytic sentences and noncircular. “Analytic,” in other words, was defined in terms of

“definition” that was defined in terms of “synonymy,” which was defined in terms of “analytic.” Thus, it was impossible to separate the truths of experience from the truths of logic and math. Later, Sellars (1956) argued that the conception of experience on which the positivists’ synthetic statements were supposed to be based was incoherent. The upshot of both criticisms was the same: there are no purely factual sentences. Every sentence, thought, and belief is part fact and part interpretation. Despite the collapse of logical positivism in the wake of these criticisms, however, many natural scientists (and philosophers) have persisted in believing that it is possible to have scientific statements that are purely factual, with no element of convention, judgment, or value. This has resulted in a distortion in both the facts these scientists claim to be studying and the values to which they oppose them.

In this context, it is interesting to note that American psychology’s “founding father” William James was aware that the separation of facts from theories and values is problematic, which led him to write approvingly of the Chicago School of psychology developed by John Dewey at the turn of the 20th century. In his short essay “The Chicago School,” James praises Dewey’s psychology for avoiding the distinction between facts and theoretical posits on one hand, and between facts and values, on the other (James, 1904, pp. 4–5).

Ecological Psychology

The style of psychology favored by James has been out of fashion for the better part of the last 50 years. In his later writings, James advocated a “radical empiricism,” according to which the objects of experience are directly perceived aspects of the world and not constructions in the mind from mental representations. On such a view, there is no division between the mind and the world, and no distinct realms of mental stuff and physical stuff. Instead of these two distinct realms, James argued, there was just one “neutral” one, which James called “pure experience.” Harry Heft (2001) has argued very convincingly that Jamesian radical empiricism forms the basis on which James Gibson’s ecological psychology was built. Gib-

son's ecological psychology, we will argue, shows that a sound experimental psychology, one whose results are replicable, cumulative, and objective, can elide the fact/value dichotomy.

Central to the ecological tradition is the concept of affordances. In Gibson's (1979) original formulation, affordances are described as opportunities for action that the environment offers or provides an animal. Gibson's description of affordances is somewhat puzzling, unless one takes it in the context of Jamesian radical empiricism:

[A]n affordance is neither an objective property nor a subjective property; or it is both if you like. An affordance cuts across the dichotomy of subjective-objective and helps us to understand its inadequacy. It is equally a fact of the environment and a fact of behavior. It is both physical and psychological, yet neither. An affordance points both ways, to the environment and to the observer (Gibson, 1979, p. 129).

Unsurprisingly given that description, understanding exactly what affordances are and how to study them empirically has proven controversial, and there are still disputes over exactly how to best understand affordances (Chemero, 2003; Chemero & Turvey, 2007; Michaels, 2000; Reed, 1996; Turvey, 1992). In what follows, we will assume that an affordance is a relation between an animal's abilities and aspects of a situation that enable those abilities.

The most famous example of this, and the first successful empirical study of affordances, is Warren (1984). Warren showed that subjects perceive the climbability of stairs as a measure of their own ability to step. In particular, he found that subjects consistently judge that the tallest step they can climb is .87 their leg length. That is, he showed that subjects perceive aspects of the environment in relation to their body scale. Warren also asked subjects to judge which riser height would be their optimally climbable step. He found that humans agree that the optimal riser height is .26 their leg length. He also found stair climbing with steps at the optimal ratio to leg length was more metabolically efficient (in terms of oxygen consumption) than steps of other heights. In other words, Warren showed that human subjects consistently and accurately perceived stair-climbing affordances as a ratio between leg length and step height. In the Warren experiments, body

scale is taken to be a measure of stepping ability. More recent research by Cesari and colleagues (2003) has quantified stair-climbing abilities more directly. They measured gait length and flexibility in both college-age and older subjects, and showed definitively that it is climbing ability and not merely body scale that subjects use to determine whether stairs are climbable.

Since Warren's initial study, there have been empirical studies of the perception of affordances relating to locomotion (Adolph, Eppler, & Gibson, 1993; Jiang & Mark, 1994; Mark, 1987; Warren & Whang, 1987), posture (Balasubramanian & Wing, 2002; Klevberg & Anderson, 2002; Riccio & Stoffregen, 1988), tool use (van Leeuwen, Smitsman, & van Leeuwen, 1994; Wagman & Carello, 2001), ball catching (Michaels, Zeinstra, & Oudejans, 2001; Oudejans, Michaels, & Bakker, 1996;), memory (Boschker, Bakker, & Michaels, 2002a, 2002b), and social relations (Clark & Uzzell, 2002; Shipley & Cohen, 2000; Valenti & Good, 1991). The concept of affordances has also become ubiquitous in human factors research, where it has been a crucial part of the development of user-centered design (Dourish, 2001; Norman, 1986). These publications, and hundreds of others, would seem to indicate that empirical studies of affordances yield results that are replicable, cumulative, and objective.

There are several features of affordances that are worth noting in this context. First, because affordances are determined in part by abilities, different animals will often (correctly) perceive different affordances in the same physical space. These will be differences between species (humans and spiders will find different things climbable), between different developmental stages within species (what is edible to an adult human will not be edible to an infant), and between different individuals of the same species (some things liftable by a body-builder will not be liftable by a concert violinist). Here, abilities include linguistic and social abilities, so there can be differences in the affordances available to members of different cultures. Second, because affordances are available in the environment for direct (nonrepresentational) and active perception, a psychology based upon affordances unifies the mind, body, and environment more thoroughly than other psychological theories. Studying affordances is studying

thought, perception, action, and the environment in which they occur simultaneously. Third, and most importantly for current purposes, affordances straddle the fact/value divide. This is because affordances are *normative*. We can see this by considering that affordances are partly constituted by abilities, and abilities are norm-governed: abilities are defined by how things are *supposed* to go, as opposed to how they actually do go; one can exercise an ability successfully or one can fail. Furthermore, as Gibson has pointed out, affordances are meanings and meanings are also normative. Indeed, Gibson's goal for his theory of affordances was to create a value-based psychology (Hodges & Baron, 1993; Reed, 1988). Affordances, then, are partly normative, yet have been the basis for high-quality scientific research. In the rest of this paper, we will argue that the scientific study of affordances can provide a model for a scientific study of virtues.

We close this section with a preliminary point. Though it is perhaps obvious, we should point out that affordances are not virtues. Humans have virtues, while affordances are in the environment (or at least in the environment-plus-human system). Thus, affordances and virtues are different kinds of things. We will address this point in detail in the next section. Although affordances are partly normative in character, the norms in question are not moral norms. There is, in other words, more than one kind of normativity and it is far from clear that the kind of normativity that infects affordances has much to do with morality. Edward Reed addressed this point indirectly in his 1996 book *Encountering the World* when he claimed claims that affordances are meanings available in the environment (Reed, 1996). The pursuit of affordances, through exploration of and learning about the environment, produces values. We can put this more clearly by saying that affordances are what makes environment *matter* to animals, and what leads them to *care* about their activity. Ecological psychology can scientifically explore questions of value and meaning, given that affordances are partly normative, but fully objective and empirically accessible.²

From Affordances to Virtues

We noted above that we are taking affordances to be relations between abilities and

behaviorally relevant aspects of situations. In the example above, the affordance being-climbable is a relation between climbing ability and height. We can define *moral analogues of affordances* along the same lines. A moral analogue of an affordance (MAA) is an opportunity for moral behavior. As with affordances, MAAs are relations between morally relevant abilities and morally relevant situations.

Take the following situation as an example: You are finishing dinner with your family when the phone rings. It is a caller from UNICEF, asking for a donation. This situation is a MAA. The MAA is a relation between your ability to provide monetary assistance to UNICEF, on the one hand, and many features of the environment, most notably the phone call and the circumstances of the children on whose behalf UNICEF is calling. As with your ability to climb, your ability to help is also multifaceted: it depends upon a series of cognitive traits (your ability to hear and understand the caller, your ability to find, and read the numbers from, a credit card) and situational traits (account balances, needed car repairs, children in private school). Most importantly, emotional, personality, and motivational traits are also involved in your ability to help the poor children that UNICEF would help: to have the ability to help, you must not be prevented from helping by hatred of foreigners, a burning desire for a yacht, or the belief that UNICEF is a wasteful, corrupt, and inefficient organization.

We suggest that the abilities that are related in moral analogues of affordances are instances of virtues. Virtues, in other words, are abilities to behave appropriately in morally relevant situations. Thus, the ability to help poor children by responding to a telephone solicitation by

² It might be argued that the kind of normativity in affordances is not the ethical kind of normativity. That is, while affordances make situations in the world matter to animals and hold value for them, there might something more (being human, being rational, being *Dasein*, etc.) to having the world matter ethically. We do not wish to take a stand on this matter in this paper. In acknowledgement of this sort of worry, though, we discuss the "moral analogues of affordances," rather than "moral affordances" in the remainder of the paper.

UNICEF is a virtue.³ We consider some implications of this understanding of virtue in the remaining part of this section.

First, the ability to help poor children by responding to a telephone solicitation by UNICEF is not what most moral philosophers and positive psychologists have in mind when they discuss virtues. Positive psychologists Christopher Peterson and Martin Seligman, for example, provide a supposedly comprehensive list of 24 different character strengths and virtues in their catalogue *Character Strengths and Virtues* (Peterson & Seligman, 2004), and the ability to help poor children by responding to a telephone solicitation from UNICEF is, unsurprisingly, not among them. Instead, they list such strengths as creativity, bravery, and kindness. We do not wish to disagree with common sense and common usage by suggesting that the ability to help poor children by responding to a telephone solicitation by UNICEF is a virtue but kindness is not. The common sense virtue term “kindness,” we suggest, is the name of a type, and the ability to help poor children by responding to a telephone solicitation by UNICEF is an instance of that type. That is the ability to help poor children by responding to a telephone solicitation by UNICEF is an instance of kindness, and kindness is a kind of virtue. Notice that this understanding of virtue makes virtues very fine-grained and situation-specific. Compare this to understanding (nonmoral) affordances and abilities. An individual might have the ability to take advantage of the affordance “legibility” when sentences are written in English but not when they are written in Japanese, or to break dance but not rumba. Similarly, a person might have the ability to respond to solicitations of funds by the American Red Cross or Oxfam but not solicitations by UNICEF, or might not be able to respond to solicitations made during dinner. Thinking of moral behavior in this manner has advantages. First, this approach makes it easy to think of individual people as being more or less kind. The more abilities like the ability to help poor children by responding to a telephone solicitation by UNICEF that one has the more kind one is. Second, these smaller, situation-specific virtues are empirically accessible in a way that common-sense virtues such as kindness are not. It is much easier to do an experiment to determine whether (and to what extent) an individual has the ability to help poor children by

responding to a telephone solicitation by UNICEF than to experiment on kindness more generally.

An empirical study of the ability to help poor children by responding to a telephone solicitation by UNICEF is indeed easy to imagine. It would, however, need to be done as part of a study of MAAs. That is, virtues must be studied along with the properties of humans that allow them to possess virtues and the environmental situations in which those virtues might become manifest. We can see how this might be done by looking again at the work by Warren (1984) and Cesari et al. (2003) on stair climbing. Warren’s study looked at one aspect of humans (length) and one aspect of environments (step height) that are involved in the affordance being-climbable. Cesari et al. improved on Warren’s experiment by showing that both leg length and flexibility are important aspects of climbing ability, and that these both affect perception of climbing affordances. In both cases, the ability, environmental situation and affordance were studied simultaneously. We can map this onto the UNICEF case as follows. The phone call from UNICEF is a moral analogue of an affordances, a relation between an ability to help and a situation in the environment. As noted above, both the ability to help and the situation in which help is called for are complex and multifaceted. We can, first of all, study to what extent individual humans have the ability to help poor children by responding to a telephone solicitation by UNICEF by putting them in situations where such help is called for, for example by calling them on the telephone, explaining the plight of poor children, and asking them to help.⁴ Then, we can collect demographic information and use standard personality instru-

³ Note that we claim that moral behavior can be *studied* objectively in this manner, not that we can objectively identify moral *edicts* in this manner. For example, many people may want to give charitably, yet would be less than keen on giving to a UN agency, given the UNs history of economic mismanagement, and instead prefer Oxfam or some other more efficient organization. Thus, they would not see this particular occasion as an opportunity for action.

⁴ It must be pointed out here that failing to give a donation in an individual instance does not *guarantee* that an individual human does not have the ability to do so. Abilities are functional, so normative, so sometimes fail. (Sometimes one mis-reads a sentence of English even in ideal conditions, yet still has the ability to read English.) A person who has the ability in question here might be in the middle of an important discussion, in a bad mood, annoyed by a call during dinner, etc.

ments on those called in the study to determine what facts about them account for most of the variance in their response to the phone call. That is, we can determine what traits of individuals make them most likely to have the virtue in question, a variety of the common-sense virtue kindness.

In recommending that positive psychologists, psychologists interested in the study of values, and empirical ethicists approach virtues as ecological psychologists approach affordances and abilities, we are also recommending the theoretical consequences of the approach. James Gibson developed his ecological psychology from James's radical empiricism, and given ecological psychology's rejection of mental representations, some ecological psychologists (e.g., Reed, 1996) have argued that the ecological tradition can be seen as a testable, empirically sound science of psychology that unifies mind and body, giving weight to facts of behavior, environment, and experience. In addition to allowing a study of virtue that is replicable, cumulative, and objective, such an approach seems to lead to a rejection of the person-situation debate in personality psychology. To summarize briefly, this debate refers to the claim that classic trait models of personality relied too heavily on "global" personality that had no predictive utility and that ignored situational influences on behavior. If one studies virtues as recommended above, as components of MAAs, virtues are only comprehensible in terms of morally relevant situations. Similarly, morally relevant situations are comprehensible only in terms of abilities to behave appropriately in them, that is, in terms of virtues. Virtues (which are traits of persons) and situations in which moral behavior occurs are interdefined. Therefore, suggesting that one or the other is responsible for explaining behavior borders on nonsense, as both the virtue and the situation are equally involved in every MAA and every action that a human takes in a MAA. Furthermore, given the fine-grained and situation-specific nature of MAAs, the virtues that one carries from situation to situation are nothing like global personality traits; rather, they are abilities to behave in very specific contexts.

Toward an Ecological Account of the Virtues

We have shown argued that affordances allow the empirical exploration of value and

meaning—they are simultaneously normative, objective, and empirically accessible. We have also argued that by following the lead of ecological psychologists' experimental work on affordances, one can study moral analogues of affordances. Doing so can make genuinely moral situations and the genuinely moral abilities (i.e., virtues) that allow humans to perceive and act upon them empirically accessible. This, it seems to us, is exactly the right way to study the "moral ecology," the environment of an individual understood in terms of moral actions that he or she might undertake in it. An ecological moral psychology can be a science of the moral ecosystem, a science that is both value-laden and fully objective.

Another advantage of an ecological approach to the study of virtues is that it would be able to render the person-situation debate irrelevant, as morally relevant situations can only be understood in terms of the capacity to behave appropriately in them—in a "virtuous" manner, that is—and conversely, virtues can only be understood in terms of situations in which they would be relevant. Seeing virtues as moral analogues of affordances would enable a more precise examination and understanding of virtuous behavior.

In addition, one would be able to study real-world situations in which individuals are expected to act virtuously. This is a significant advantage over contemporary methods in both positive psychology and experimental ethics. For example, Peterson and Seligman's survey of an individual's "signature strengths" only measures subjective individual assessments of one's perceived strengths and virtues. It seems unlikely that we would be able to gain real insight into how humans actually act solely through self-report measures. Of course, this is an issue that resonates in many areas of personality and social psychology, and is therefore not an issue specific to positive psychology. By putting experimental participants in positions in which they have the opportunity to put their virtues into action, one avoids the problems attendant to self-reports of character strengths.

The recommended perspective and methodology will be helpful to experimental ethicists in two ways. First, it will allow them to develop data-driven explanations of individual virtues and character strengths in a manner similar to prior work on concepts such as purposefulness

(Knobe, 2003) and freedom (Nichols, 2004; Nahmias et al., to appear). Second, and perhaps more importantly, it will allow empirical ethicists to move beyond studying abstract concepts such as these, and will give them the opportunity to study moral behavior, the abilities that allow humans to engage in moral behavior, and the circumstances in which moral behavior is possible.

We have argued that a Gibsonian approach toward the study of human strengths holds many advantages. The next step would be to perform experiments that put this conceptualization to the test. Such a step is soon forthcoming.

References

- Adolph, K., Eppler, M., & Gibson, E. (1993). Crawling versus walking infants' perception of affordances for locomotion over sloping surfaces. *Child Development*, 64, 1158–1174.
- Balasubramanian, R., & Wing, A. M. (2002). The dynamics of standing balance. *Trends in Cognitive Sciences*, 6, 531–536.
- Boschker, M., Bakker, F., & Michaels, C. (2002a). Memory for the functional characteristics of climbing walls: Perceiving affordances. *Journal of Motor Behavior*, 34, 25–36.
- Boschker, M., Bakker, F., & Michaels, C. (2002b). Effect of mental imagery on realizing affordances. *Quarterly Journal of Experimental Psychology: Human Experimental Psychology*, 55A, 775–792.
- Cesari, P., Formenti, F., & Olivato, P. (2003). A common perceptual parameter for stair climbing for children, young and old adults. *Human Movement Science*, 22, 111–124.
- Chemero, A. (2003). An outline of a theory of affordances. *Ecological Psychology*, 15, 181–195.
- Chemero, A., & Turvey, M. T. (2007). Hypersets, complexity, and the ecological approach to perception-action. *Biological Theory*, 2, 23–26.
- Clark, C., & Uzzell, D. (2002). The affordances of the home, neighborhood, school and town centre for adolescents. *Journal of Environmental Psychology*, 22, 95–108.
- Csikszentmihalyi, M. (2003). Legs or wings? A reply to R. S. Lazarus. *Psychological Inquiry*, 14, 113–115.
- Dewey, J. (1900). Psychology and social practice. *Psychological Review*, 7, 121.
- Dousrith, P. (2001). *Where the action is*. Cambridge: MIT Press.
- Gibson, J. J. (1979). *The ecological approach to visual perception*. Boston: Houghton Mifflin.
- Heft, H. (2001). *Ecological psychology in context: James Gibson, Roger Barker, and the legacy of William James's radical empiricism*. Mahwah, NJ: Erlbaum, Inc.
- Hodge, B. H., & Baron, R. M. (1992). Values as constraints on affordances- perceiving and acting properly. *Journal for the Theory of Social Behavior*, 22, 263–294.
- James, W. (1976). *Essays in radical empiricism*. Cambridge, MA: Harvard University Press. (Original work published 1912).
- Jiang, Y., & Mark, L. (1994). The effect of gap depth on the perception of whether a gap is crossable. *Perception & Psychophysics*, 56, 691–700.
- Kendler, H. (1999). The role of value in the world of psychology. *American Psychologist*, 54, 828–835.
- Klevberg, G., & Anderson, D. (2002). Visual and haptic perception of postural affordances in children and adults. *Human Movement Science*, 21, 169–186.
- Knobe, J. (2003). Intentional action in folk psychology: An experimental investigation. *Philosophical Psychology*, 16, 309–324.
- Mark, L. (1987). Eyeheight-scaled information about affordances: A study of sitting and stair climbing. *Journal of Experimental Psychology: Human Perception and Performance*, 13, 361–370.
- Michaels, C. F. (2000). Information, perception, and action: What should ecological psychologists learn from Milner and Goodale (1995)? *Ecological Psychology*, 12, 241–258.
- Michaels, C. F., Zeinstra, E. B., & Oudejans, R. R. D. (2001). Information and action in punching a falling ball. *Quarterly Journal of Experimental Psychology*, 54, 69–93.
- Nahmias, E., Morris, S. G., Nadelhoffer, T., & Turner, J. (2006). Is incompatibilism intuitive? *Philosophy and Phenomenological Research*, 73, 28–53.
- Nichols, S. (2004b). Folk psychology of free will. *Mind & Language*, 19, 473–502.
- Norman, D. (1986). *The psychology of everyday things*. New York: Basic Books.
- Oudejans, R., Michaels, C., & Bakker, F. (1996). The relevance of action in perceiving affordances: Perception of catchableness of fly balls. *Journal of Experimental Psychology: Human Perception and Performance*, 22, 879–891.
- Peterson, C., & Seligman, M. E. P. (2004). *Character strengths and virtues*. Oxford: Oxford University Press.
- Quine, W. V. (1953). Two dogmas of empiricism. In Quine, W. V. (1953). *From a logical point of view*. Cambridge: Harvard University Press.
- Reed, E. S. (1996). *Encountering the world*. New York: Oxford University Press.
- Reed, E. S. (1988). *James J. Gibson and the psychology of perception*. New Haven, CT: Yale University Press.

- Riccio, G., & Stoffregen, T. (1988). Affordances as constraints on the control of stance. *Human Movement Science*, 7, 2–4.
- Schwartz, B. (2000). Self-determination: The tyranny of freedom. *American Psychologist*, 55, 79–88.
- Seligman, M. E. P. (2002). *Authentic happiness*. New York: The Free Press.
- Seligman, M. E. P. (2002). Positive psychology, positive prevention, and positive therapy. In C. R. Snyder & S. J. Lopez (Eds.), *The handbook of positive psychology* (pp. 3–12). New York: Oxford Press.
- Seligman, M. E. P., & Csikszentmihalyi, M. (2000). Positive psychology: An introduction. *American Psychologist*, 55, 1–20.
- Seligman, M. E. P., & Csikszentmihalyi, M. (2001). Reply to comments. *American Psychologist*, 56, 89–90.
- Sellars, W. (1956). Empiricism and the philosophy of mind. In H. Feigl, & M. Scriven (Eds.), *Minnesota studies in the philosophy of science* (Vol. I). Minneapolis: University of Minnesota Press.
- Shipley, T., & Cohen, L. (2000). Affordances for coordinated action in point-light walker displays. *Ecological Psychology*, 12, 87–92.
- Turvey, M. (1992). Affordances and prospective control: An outline of the ontology. *Ecological Psychology*, 4, 173–187.
- Valenti, S., & Good, J. (1991). Social affordances and interaction: I. Introduction. *Ecological Psychology*, 3, 77–98.
- van Leeuwen, L., Smitsman, A., & van Leeuwen, C. (1994). Affordances, perceptual complexity, and the development of tool use. *Journal of Experimental Psychology: Human Perception and Performance*, 20, 174–191.
- Wagman, J., & Carello, C. (2001). Affordances and inertial constraints on tool use. *Ecological Psychology*, 13, 173–195.
- Warren, W. H. (1984). Perceiving affordances: Visual guidance of stair climbing. *Journal of Experimental Psychology: Human Perception and Performance*, 10, 683–703.
- Warren, W. H., & Whang, S. (1987). Visual guidance of walking through apertures: Body-scaled information for affordances. *Journal of Experimental Psychology: Human Perception and Performance*, 13, 371–383.

Received February 1, 2007

Accepted February 1, 2008 ■